

APPLIED NATURAL LANGUAGE PROCESSING • UNIVERSITY OF TRENTO • MAY 2026

Summarising Football Match Commentaries

A comparative study of prompting and fine-tuning on long-form sports audio transcripts

Marco Panciera • Christian Dalfarra

2.5 hours of radio → a 200-word match report

- ▶ **Input:** Whisper transcript of live BBC Radio 5 commentary — mean ~25,000 words, noisy, fragmented, full of irrelevant detail.
- ▶ **Output:** a structured match report (~200 words): score line, key events, short tactical note.
- ▶ **Why it is hard:** inputs exceed every standard transformer context window (1,024 tokens ≈ 4% of a match); references for training are scarce.
- ▶ **Two empirically answerable questions**
 1. Chunk-and-aggregate, or a long-context encoder (LED-16384)?
 2. For each input strategy — prompting a general model, or fine-tuning on small in-domain data?

99 matches, 43 years, 12 competition categories

BBC radio audio → Whisper transcription → GPT-4 reference summaries

- ▶ **Audio:** BBC Radio 5 Live archive (YouTube, via yt-dlp) — ~19 GB MP3, matches from 1983 to 2026.
- ▶ **Transcription:** Whisper-small; full transcript + time-aligned segments with pitch/energy prosodic features.
- ▶ **References:** GPT-4 (ChatGPT web, search enabled) prompted per match identifier; facts grounded in web-retrieved match reports, BBC sourcing verified for every match; fixed template, ~190 words. A pragmatic, disclosed compromise — no human summaries exist for radio commentary.
- ▶ **Split:** 80 train / 10 val / 9 test, stratified, committed to the repo for reproducibility.
- ▶ **Known noise floor:** Whisper systematically corrupts player names (“Bergkamp” → “Burgkamp”) — an irreducible ceiling on surface-overlap metrics.

2 input pipelines × 2 learning paradigms

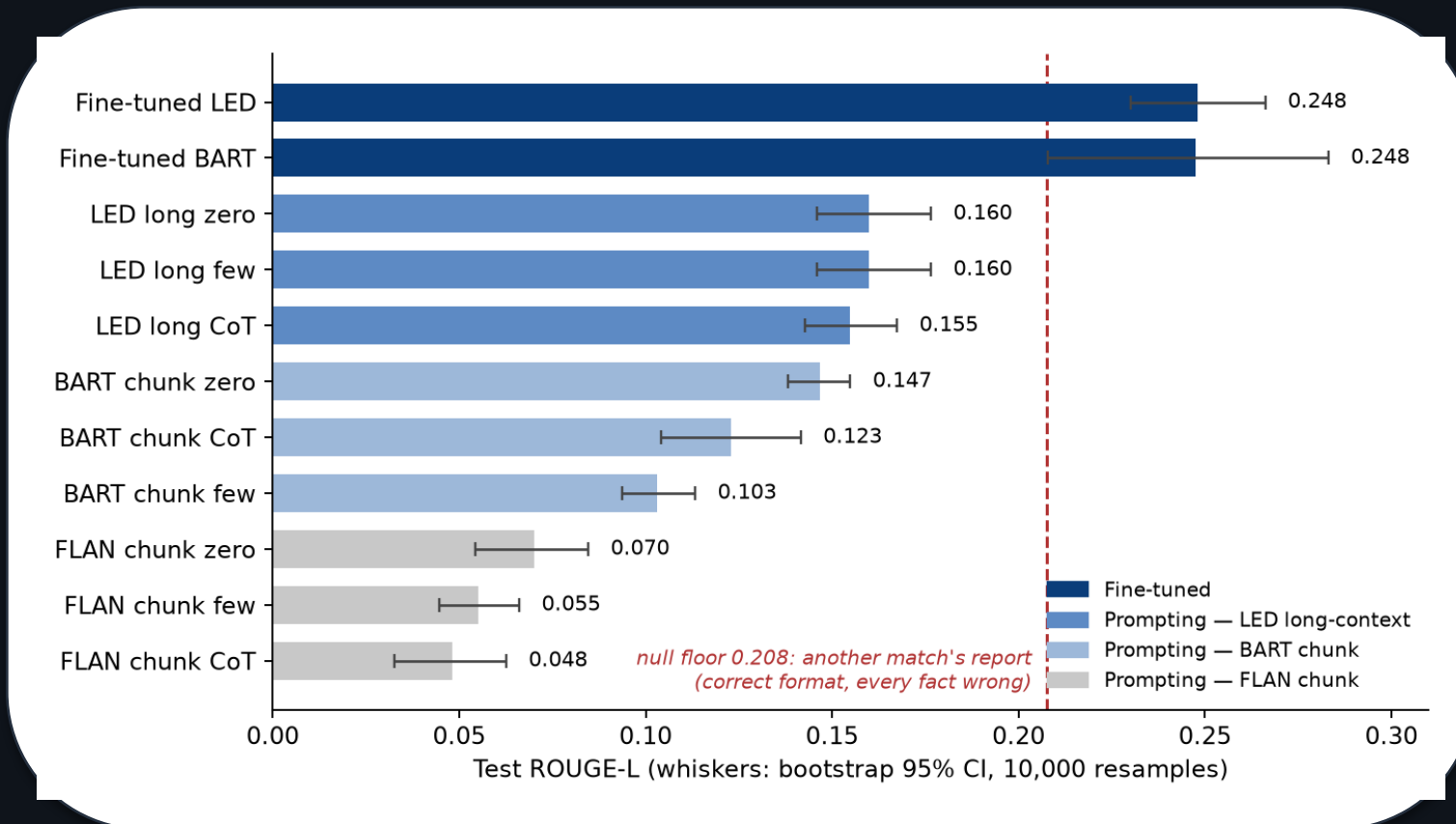
3 architectures · 3 prompt styles · 11 evaluated conditions

	Chunk + aggregate	Long-context (16,384 tokens)
Prompting (zero / few / CoT)	FLAN-T5-Large (512) BART-Large-CNN (1,024)	LED-base-16384
Fine-tuning	BART as the merge step ("chunk-merger" recipe)	LED on 8,192-token random windows

- ▶ **Chunk + aggregate:** segment-aligned chunks → per-chunk summary → merge + dedup. **Long-context:** whole transcript in one LED pass.
- ▶ All experiments on free-tier Kaggle (1× T4); seeds fixed; every prediction saved to disk, so all metrics are recomputable post-hoc.

Fine-tuning wins — but only it clears the null floor

Test ROUGE-L on the clean 9-match test set · bootstrap 95% CIs (10,000 resamples)



Every prompting condition is significantly below both fine-tuned models (paired bootstrap; 95% CI of the difference excludes 0) — and below the dashed null floor. Only the fine-tuned models clear it, by +0.040 [+0.022, +0.059]. The LED-BART gap (−0.0004 [−0.042, +0.035]) is NOT significant.

Match the training distribution, not the context window

The “chunk-merger” fine-tuning recipe

- ▶ **Naive fine-tuning** on (transcript[:1024] → reference) shows BART only the first ~5 minutes of a 2.5-hour match.
- ▶ **Chunk-merger recipe:** run pre-trained BART on every chunk → concatenate the ~30 mini-summaries (~960 tokens) → fine-tune BART to map that intermediate to the reference report. Training input now matches what the model sees at inference.
- ▶ **Two-phase schedule:** encoder frozen 3 epochs, then unfrozen; early stopping on val ROUGE-L.
- ▶ **Same model, same pipeline, same test set**

pre-trained BART 0.147 → fine-tuned BART 0.248 (+69% relative)

+55% over the strongest prompting condition of any architecture (LED zero-shot, 0.160) — but that baseline sits below the null floor; see Finding 3.

Long context wins at prompting, ties at fine-tuning

- ▶ **Prompting:** LED long-context (avg 0.158) beats BART chunk (0.124) and FLAN chunk (0.058) — the 16k window preserves cross-segment information that chunking discards.
- ▶ **Fine-tuning:** LED long-context is nominally ahead (0.248 vs 0.248) but the paired difference is not significant at $n = 9$ — the two fine-tuned pipelines cannot be separated. LED is more consistent (CI [0.230, 0.266]); BART has more variance (CI [0.208, 0.283]).
- ▶ **Why the two pipelines are statistically tied despite architectural differences:**
 - BART learns from a ~960-token pre-distilled intermediate whose features are already informative.
 - LED must learn what matters inside raw 8,192-token windows from scratch — too few pairs ($N = 80$) to learn long-range attention patterns reliably.
- ▶ **Secondary finding:** zero-shot beats few-shot and CoT in both chunked families — examples eat the input budget; LED few-shot fell back to zero-shot (implementation gap, disclosed in the report), and LED-base is not instruction-tuned, so prompt style barely matters.

Models learn the format, not the facts

Real test-set outputs; each quote labelled with the true result

Fine-tuned BART (ROUGE rank 2) — 2019 UCL Final (true: Liverpool 2-0 Tottenham, Madrid, no shootout)

“Liverpool 2-2 Tottenham Hotspur (Liverpool win 4-3 on penalties) ... 2023 UEFA Champions League Final, Wembley Stadium, London – 31 May 2019 ... Liverpool secured a dramatic victory...”

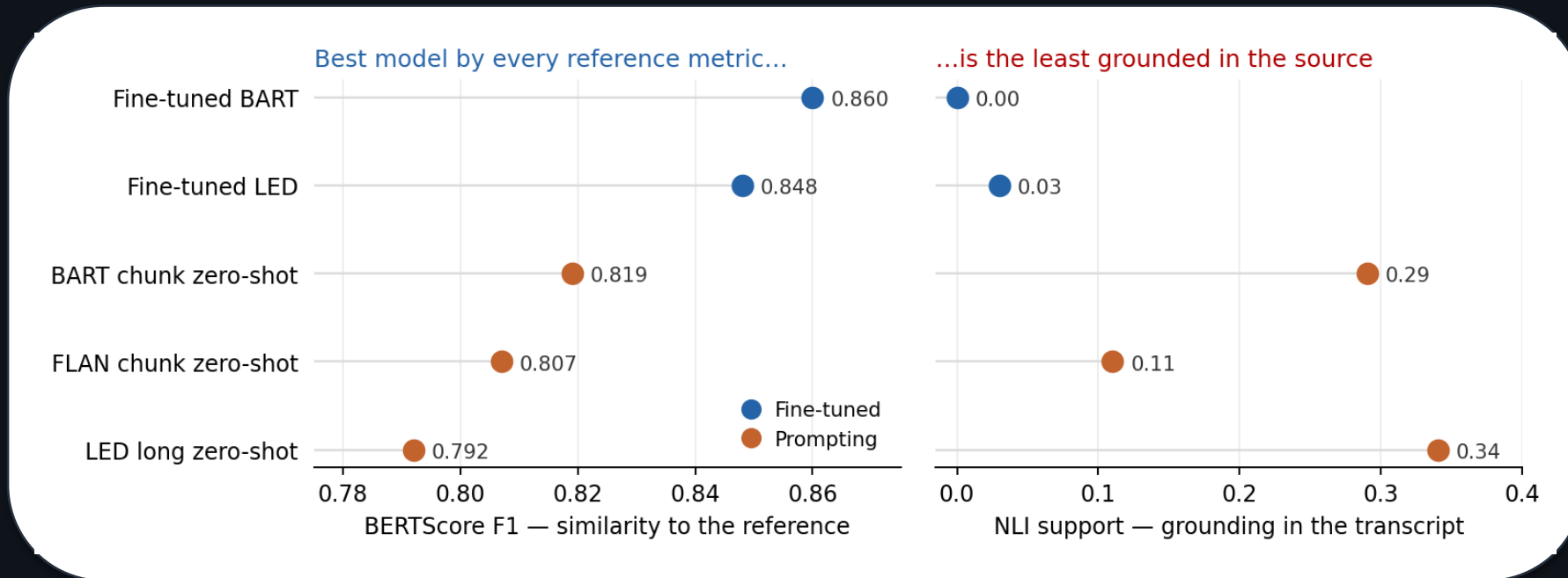
Fine-tuned LED (ROUGE rank 1) — 2001 FA Cup Final (true: Arsenal 1-2 Liverpool, Millennium Stadium)

“Liverpool 1-0 Arsenal · 2023-24 UEFA Cup Final, Millennium Stadium, Cardiff, UK – 24 May 2021 ... Mohamed Salah curled a low shot into the top corner...” — wrong score, wrong date, wrong competition, scorers who were toddlers in 2001; the venue is the one grounded detail.

- ▶ **Quantified over all 9 test matches:** fine-tuned BART (ROUGE rank 2) states the correct final score in **0 of 9 matches** and recalls only 30% of reference scorers. Fine-tuned LED (ROUGE rank 1): 2/9 scores, 13% scorers. BART scorelines are decorative — 1-1, 2-2, 1-0 regardless of the real 0-4, 3-0...
- ▶ **ROUGE rewards the format, and we measured how much:** a wrong match’s report — right template, every fact wrong — already scores ROUGE-L 0.208, 84% of our best model. Every prompting condition scores below a template with no match facts at all. Only +0.040 [+0.022, +0.059] of the fine-tuned score is real match content.
- ▶ **And nothing was lost in transcription:** all 22 reference scorers appear in the transcripts the models read — a 100% ceiling on scorer recall, of which BART realises 30% and LED 13%. Extraction failure, not ASR noise.

Models optimise the reference, not the source

BERTScore vs reference • NLI entailment vs transcript (DeBERTa-MNLI, TF-IDF-retrieved windows)



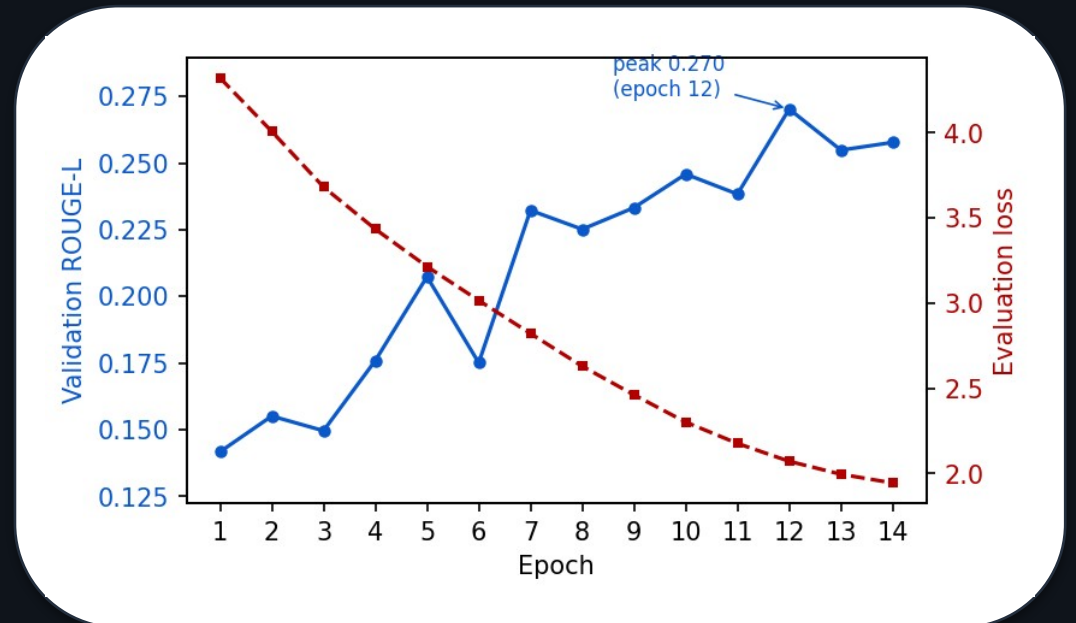
► **The rankings invert:** fine-tuned LED leads on ROUGE-L and BERTScore yet has only 2% NLI support; fine-tuned BART — zero NLI-supported sentences — is even more extreme. Prompted long-context LED, outranked on every reference metric, is the best-grounded condition (34%) because it copies the commentary.

► **Fine-tuning on GPT-4 references optimises agreement with the references at the expense of grounding in the input.** (NLI rates are conservative lower bounds — register gap, ASR noise — so compare across conditions, not absolutely.)

The biggest improvement was not a model change

Five silent LED training failures + one dataset bug

- ▶ **33% of training references were 0-byte files** (and 44% of test references) — silently teaching models to emit empty strings and deflating every score $\sim 1.8\times$.
- ▶ Predicted correction ratio matched the measured one almost exactly (e.g. fine-tuned BART $0.136 \rightarrow 0.248$ after the fix).
- ▶ **LED also needed 5 bug fixes** (defeated random-window sampler, collapse-to-EOS, decoder sentinel overflow, lost checkpoint, CLI truncation).
- ▶ **After the fixes (run 5, clean data): 14 epochs, val ROUGE-L peaks at 0.270 $\rightarrow$ test 0.248. Earlier collapses were data and code defects, not modelling limits.**



LED run 5 (clean dataset): val ROUGE-L peaks at 0.270 (epoch 12) $\rightarrow$ test 0.248.

What we learned, and where it breaks

- ▶ **1. Fine-tuning beats prompting at small N; the two fine-tuned pipelines tie** — +69% over the same pre-trained model under the same pipeline, +55% over the best prompting baseline; against a format-matched null the honest margin is +0.040 ROUGE-L [+0.022, +0.059]. LED and BART are statistically indistinguishable at $n = 9$.
- ▶ **2. Context length is not a substitute for supervision** — long-context wins only when no fine-tuning is possible.
- ▶ **3. Reference similarity and source grounding fully decouple** — fine-tuned LED leads every reference metric yet gets 2/9 scores and 2% NLI; BART is worse at 0/9 and 0% NLI. The prompted LED both outrank on ROUGE is the best-grounded condition (34%). Optimising faithfulness, not just measuring it, is the natural next step.

▶ Limitations we state openly

Search-grounded GPT-4 references → facts anchored in web reports, but still ungrounded in the transcript, stylistically homogeneous, and not exactly reproducible (search results drift).

$n = 9$ test matches → bootstrap CIs resolve fine-tuning vs prompting (significant) but not LED vs BART (-0.0004 [$-0.042, +0.035$]).

$N = 80$ training pairs → both fine-tuned models overfit (val 0.31 → test 0.25 for BART).

NLI support rates are conservative lower bounds (ASR-noisy premises, register gap).

Reference metrics are format-saturated — 84% of our ROUGE-L needs no correct facts.